



13.9.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson?



Unit 13, Lesson 10: Determining Which Function Type Best Models a Real-World Context



Benchmark

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.



Learning Target

- ☐ I can determine whether a linear, quadratic, or exponential function best models a given real-world situation.

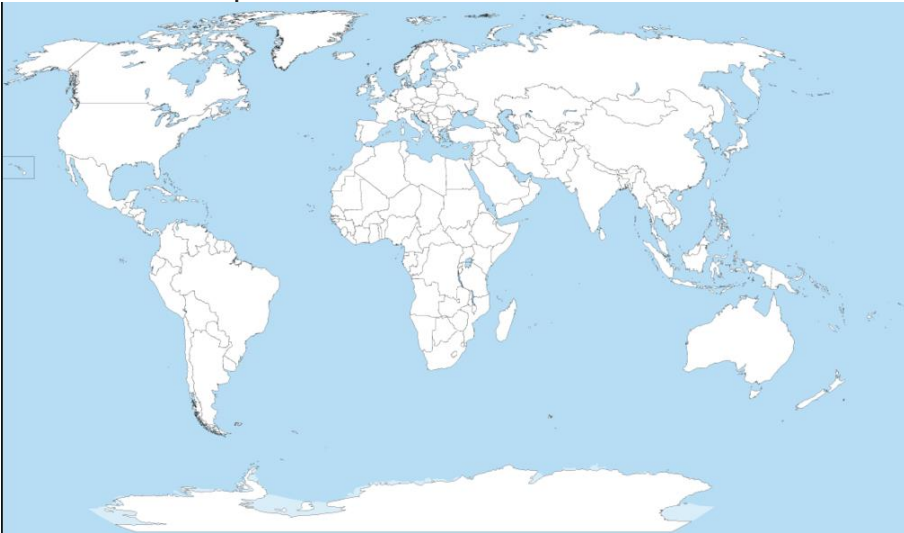


13.10.1 Warm-Up: International Poverty Line

1. The International Poverty Line is the base that determines whether someone is living in poverty. In 2021, the International Poverty Line was \$1.90 per day.

A typical year contains 261 working days. What is the yearly salary for someone living at the International Poverty Line?

2. A world map is shown.



Place a dot on the map to represent as many of the following countries as you can.

Bangladesh	Ghana	Rwanda
China	Ireland	Spain
Congo	Peru	United Kingdom
Ethiopia	Romania	



13.10.2 Activity: Card Sort

1. Sort the cards to match each function type with the table of values, graph, and written description.

Function Type	Table of Values	Graph	Written Description
Linear			
Quadratic			
Exponential			

- a. Describe the strategies you used in the card sort.
- b. Discuss with your partner the strategies that can be used to determine the type of function from the table of values. Summarize your discussion.
- c. Discuss with your partner what strategies can be used to determine the type of function from the written description. Summarize your discussion.



13.10.3 Activity: Gallery Walk

Each table or written description gives information about the percentage of people living at or below the International Poverty Line for the given time period, where 0 represents the first year in the given time period.

- Use the provided information to determine whether a linear, quadratic, or exponential function best models the table or written description.
- On your paper, include the function type and a justification for your choice.
- If your group has a table of values, then create a written description.
- If your group has a written description, then create a table of values.

During the gallery walk record the following information.

Country	Function Type	Useful Information to Determine Function Type
Bangladesh		
China		
Congo		
Ethiopia		
Ghana		
Peru		
Romania		
Rwanda		



13.10.4 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found multiple strategies	☐	☐
I persevered	☐	☐
Eliminated negative self-talk	☐	☐